

# Braking System Requirements Specification

## 1. Scope

This document specifies synthetic but realistic requirements for automotive braking systems, covering both legacy hydraulic architectures and modern brake-by-wire one-box electro-hydraulic architectures. The scope includes service braking, parking braking, ABS, ESC, EPB, regenerative braking coordination, diagnostics, degraded modes, and functional safety requirements aligned with common automotive braking and safety practices.[web:6][web:7][web:17][web:22]

The document is written as a pure requirements specification. It excludes RAG ingestion guidance, chunking strategy, retrieval examples, and corpus-structuring content that is not part of the braking system specification itself.[file:32][file:33]

## 2. System Overview

A legacy hydraulic brake system uses a brake pedal, booster, master cylinder, hydraulic lines, and wheel-end friction brakes to convert driver input into braking torque. ABS and ESC functions are typically implemented through an ECU and hydraulic modulation components connected into the hydraulic circuit.[web:13][web:14][web:15][web:30]

A brake-by-wire one-box system replaces or decouples the direct mechanical/hydraulic pedal-to-wheel path and interprets driver demand through sensors, electronic control logic, and an integrated electro-hydraulic actuator unit. Such systems commonly integrate service braking, ABS, ESC, traction control, and coordination with regenerative braking in electrified vehicles. [web:18][web:21][web:23][web:28]

## 3. Architectural Description

### 3.1 Legacy Hydraulic Architecture

The legacy architecture shall comprise the following primary elements:[web:13][web:14][web:15]

- Brake pedal and mechanical linkage.
- Vacuum or hydraulic brake booster.
- Tandem master cylinder.
- Dual hydraulic circuits.
- Wheel-end disc or drum brake actuators.
- Wheel speed sensing and hydraulic modulation components for ABS/ESC-equipped variants.

## 3.2 Brake-by-Wire One-Box Architecture

The one-box brake-by-wire architecture shall comprise the following primary elements:  
[web:18][web:21][web:28]

- Brake pedal module with redundant sensing.
- Pedal feel simulator.
- Central brake ECU or integrated control electronics.
- Electro-hydraulic pressure generation unit.
- Individually controlled hydraulic output paths for wheel brakes.
- Pressure, wheel-speed, and system-status sensing.
- Interfaces to ADAS, powertrain, and vehicle dynamics controllers.

## 4. Definitions

- **Service Brake:** Primary braking function used to decelerate or stop the vehicle during driving.[web:7]
- **Parking Brake:** Function intended to hold the vehicle stationary under specified loading and gradient conditions.[web:7]
- **ABS:** Anti-lock Braking System used to prevent wheel lock during braking.[web:6][web:8]
- **ESC/ESP:** Electronic Stability Control function used to stabilize vehicle yaw behavior through selective braking interventions.[web:30][web:6]
- **EPB:** Electric Parking Brake.[web:18][web:28]
- **BBW:** Brake-by-wire braking architecture using electrical/electronic interpretation of brake demand and electronically commanded brake actuation.[web:23][web:28]
- **EHB:** Electro-hydraulic braking implementation of brake-by-wire.[web:18][web:28]
- **HARA:** Hazard Analysis and Risk Assessment within the ISO 26262 safety lifecycle.[web:17][web:22]
- **ASIL:** Automotive Safety Integrity Level defined by ISO 26262.[web:17][web:22]

## 5. System-Level Requirements

### REQ-SYS-001 – Service Brake Provision

The vehicle shall provide a service brake system capable of controlled deceleration and stopping across the full operating range of the vehicle, including specified laden and unladen conditions.[web:7][web:11]

### REQ-SYS-002 – Parking Brake Provision

The vehicle shall provide a parking brake function capable of holding the vehicle stationary under specified gradient and load conditions in both forward and reverse directions.[web:7][web:11]

### REQ-SYS-003 – Stability Support Functions

Where equipped, the braking system shall support ABS and ESC functions without

compromising the availability of the base braking function.[web:6][web:30]

#### **REQ-SYS-004 – Residual Braking**

The braking system shall provide residual braking capability under defined single-point failure conditions so that total loss of braking does not occur after a single failure in the service brake path.[web:7][web:25]

#### **REQ-SYS-005 – Functional Safety Lifecycle**

Electrical and electronic braking functions shall be developed in accordance with the ISO 26262 functional safety lifecycle, including item definition, HARA, safety goals, and derived technical safety requirements.[web:17][web:22]

## **6. Legacy Hydraulic Requirements**

### **6.1 Functional Requirements**

#### **REQ-LG-001 – Hydraulic Service Braking**

The legacy braking system shall use hydraulic pressure generated by a master cylinder to actuate wheel-end friction brakes.[web:13][web:15]

#### **REQ-LG-002 – Dual Circuit Design**

The service brake shall use at least two hydraulic circuits arranged such that loss of one circuit does not eliminate all service braking capability.[web:7][web:25]

#### **REQ-LG-003 – Brake Booster Assistance**

The legacy system shall use a vacuum booster or hydraulic booster to reduce required pedal effort during normal braking.[web:14][web:15][web:5]

#### **REQ-LG-004 – Proportional Pedal Response**

Pedal force and travel shall produce predictable and approximately proportional hydraulic braking response over the normal operating range.[web:5][web:13]

#### **REQ-LG-005 – Pressure Generation Capacity**

The master cylinder and booster combination shall generate sufficient pressure to meet required braking performance under worst-case loading and temperature conditions.[web:5][web:7]

#### **REQ-LG-006 – Booster Failure Operation**

Loss of booster assistance shall not prevent service braking; the driver shall retain the ability to achieve a defined minimum braking performance using increased pedal effort.[web:5][web:15]

#### **REQ-LG-007 – Hydraulic Failure Warning**

The system shall warn the driver upon detection of significant hydraulic circuit degradation, pressure loss, or brake fluid loss affecting braking performance.[web:7][web:25]

#### **REQ-LG-008 – ABS/ESC Compatibility**

In ABS- or ESC-equipped variants, the hydraulic architecture shall permit modulation by valves and control logic without degrading baseline hydraulic brake operation when electronic functions are unavailable.[web:6][web:30]

## 6.2 Performance and Robustness Requirements

### REQ-LG-020 – Brake Fade Resistance

The legacy system shall maintain required braking performance after repeated high-energy braking events within defined thermal operating limits.[web:6][web:9]

### REQ-LG-021 – Environmental Robustness

The hydraulic braking system shall maintain functional performance under environmental exposure to water, dust, road contaminants, and expected temperature extremes.[web:6][web:21]

### REQ-LG-022 – Parking Brake Retention

The legacy parking brake shall provide a mechanical or mechanically retained holding function such that loss of hydraulic pressure or electrical supply does not release the vehicle unintentionally.[web:7][web:10]

## 7. ABS and ESC Requirements

### REQ-ABS-001 – Wheel Speed Acquisition

The braking control system shall acquire wheel speed information from all instrumented wheels at a sampling rate sufficient to detect impending lock-up in time for closed-loop pressure modulation.[file:32][web:6]

### REQ-ABS-002 – Anti-Lock Pressure Modulation

When wheel slip exceeds control thresholds, the ABS function shall modulate individual wheel brake pressure to reduce lock-up risk and preserve steerability.[web:6][web:30]

### REQ-ABS-003 – Wheel Speed Sensor Diagnostics

The system shall continuously monitor wheel speed sensor electrical plausibility and signal integrity and shall detect open-circuit, short-circuit, stuck, and implausible sensor conditions.[file:32][web:25]

### REQ-ABS-004 – ABS Degraded Operation

Upon detection of an ABS fault that prevents reliable anti-lock control, the system shall disable ABS interventions and maintain conventional base braking capability while issuing a driver warning.[file:32][web:25]

### REQ-ESC-001 – Vehicle Stability Control

The ESC function shall selectively apply wheel brake interventions to mitigate unstable yaw behavior when sensed vehicle motion deviates beyond calibrated stability thresholds.[web:30][file:32]

### REQ-ESC-002 – Brake and Torque Coordination

The ESC function shall coordinate wheel braking requests with powertrain torque reduction requests where such interface is available.[file:32][web:30]

### REQ-ESC-003 – ESC Fault Handling

Loss of ESC function shall not degrade the base service brake function and shall trigger an appropriate driver indication.[web:25][file:32]

## 8. Electric Parking Brake Requirements

### REQ-EPB-001 – Electric Parking Actuation

Where an EPB is provided, the parking brake shall be actuated by an electric mechanism capable of applying and holding the brake under specified load and gradient conditions.  
[web:18][web:28]

### REQ-EPB-002 – Overcurrent Protection

The EPB actuator control shall detect sustained motor overcurrent or overload conditions and transition to a fault state to prevent thermal or mechanical damage.[file:32]

### REQ-EPB-003 – Status Monitoring

The EPB function shall monitor actuator position, current, or equivalent state variables to confirm successful apply and release actions.[file:32]

### REQ-EPB-004 – Fault Indication

Failure of the EPB to apply, release, or maintain holding force shall be indicated to the driver and logged by diagnostics.[file:32]

## 9. Brake-by-Wire One-Box Requirements

### 9.1 Functional Requirements

#### REQ-BBW-001 – Decoupled Brake Demand Sensing

The one-box brake-by-wire system shall interpret driver brake demand using redundant electrical sensing of pedal position, force, or equivalent pedal input variables.[web:18][web:23][web:28]

#### REQ-BBW-002 – Pedal Feel Simulation

The system shall provide a pedal feel simulator that reproduces stable and repeatable pedal feedback independent of direct wheel-circuit hydraulic pressure.[web:18][web:28]

#### REQ-BBW-003 – Electro-Hydraulic Pressure Generation

The one-box unit shall generate brake pressure electro-hydraulically and distribute commanded pressure to the wheel brakes without requiring a conventional vacuum booster for normal operation.[web:18][web:21][web:28]

#### REQ-BBW-004 – Integrated Dynamics Functions

The one-box architecture shall support integrated ABS, ESC, and traction-related brake interventions through a shared electro-hydraulic unit.[web:18][web:21][web:30]

#### REQ-BBW-005 – Full Vehicle Braking Capability

The brake-by-wire system shall generate sufficient brake pressure on all controlled wheels to satisfy vehicle-level stopping and stability requirements under normal operating conditions.  
[web:18][web:28]

#### REQ-BBW-006 – Redundant Pressure Path or Channel Strategy

The one-box braking architecture shall implement redundancy or equivalent fault-tolerant

measures such that a single failure in sensing, actuation, or pressure generation does not immediately result in total loss of service braking.[file:32][web:23][web:25]

#### **REQ-BBW-007 – Autonomous Brake Request Acceptance**

The brake ECU shall accept braking requests from authorized ADAS functions, including AEB, subject to safety arbitration and brake system availability.[web:12][web:29]

#### **REQ-BBW-008 – Safe Degraded Operation**

The one-box system shall detect internal faults and enter a degraded operating state that preserves a defined minimum braking capability and prevents hazardous unintended braking.[web:16][web:23][web:25]

## **9.2 Detailed Control Requirements**

#### **REQ-BBW-020 – Redundant Pedal Sensor Plausibility**

Redundant pedal sensor channels shall be cross-checked for plausibility, and discrepancies beyond calibrated limits shall trigger a safe-state or degraded-mode response.[file:32][web:23]

#### **REQ-BBW-021 – Wheel Pressure Control**

The one-box unit shall independently control hydraulic pressure at each controlled wheel to support anti-lock braking, stability control, and torque-vectoring-related brake interventions where applicable.[web:21][web:30]

#### **REQ-BBW-022 – Demand-to-Deceleration Mapping**

During normal operation, the brake-by-wire control software shall map driver brake demand to target deceleration and wheel pressure commands with stable and predictable vehicle response.[web:18][web:28]

#### **REQ-BBW-023 – Regenerative Braking Coordination**

In electrified vehicles, the braking controller shall coordinate regenerative and friction braking so that total deceleration tracks driver demand while maintaining stability, brake feel consistency, and fallback braking capability.[web:18][web:28]

#### **REQ-BBW-024 – Low-Speed Blending Transition**

The braking controller shall manage regenerative-to-friction blending transitions, particularly at low vehicle speeds and high deceleration requests, to avoid perceivable braking discontinuities beyond calibrated limits.[web:28][file:33]

## **10. Diagnostic Requirements**

#### **REQ-DIAG-001 – Fault Detection Coverage**

The braking system shall detect faults affecting sensors, actuators, communication, power supply, and pressure generation that could compromise braking performance or safety functions.[web:16][web:25]

#### **REQ-DIAG-002 – Diagnostic Trouble Codes**

Detected faults shall be recorded using structured diagnostic trouble code entries or equivalent diagnostic identifiers that uniquely identify the failed subsystem or failure mode.[file:32]

#### **REQ-DIAG-003 – Startup Self-Test**

The brake control electronics shall perform startup self-tests on critical sensing, actuation, memory, and monitoring functions prior to enabling normal control.[file:32][web:16]

#### **REQ-DIAG-004 – Continuous Runtime Monitoring**

The braking controller shall perform runtime monitoring of watchdog status, communication integrity, power supply validity, sensor plausibility, and actuator response.[file:32][web:16]

#### **REQ-DIAG-005 – Fault Memory Storage**

Upon occurrence of a safety-relevant fault, the system shall store sufficient diagnostic context in non-volatile memory to support post-event analysis and service diagnostics.[file:32]

#### **REQ-DIAG-006 – Driver Warning Strategy**

Faults that affect braking performance, safety functions, or parking brake availability shall be communicated to the driver through tell-tales, messages, or equivalent HMI means.[web:7][web:25][file:32]

### **11. Functional Safety Requirements**

#### **REQ-FS-001 – Safety Goal Definition**

Vehicle-level safety goals shall be defined for hazards including loss of braking, insufficient braking, unintended braking, asymmetric braking, and loss of directional stability during braking.[web:16][web:24][web:25]

#### **REQ-FS-002 – HARA Execution**

A Hazard Analysis and Risk Assessment shall be performed for all safety-relevant braking functions, including legacy electronically controlled subfunctions and brake-by-wire functions.[web:17][web:22][web:23]

#### **REQ-FS-003 – ASIL Allocation**

Safety goals and derived safety requirements shall be assigned ASIL classifications in accordance with ISO 26262 based on severity, exposure, and controllability.[web:17][web:22]

#### **REQ-FS-004 – Technical Safety Requirements**

System, hardware, and software technical safety requirements shall be derived from the safety goals and allocated to the relevant architectural elements.[web:16][web:22][web:25]

#### **REQ-FS-005 – Safe State Definition**

For each identified safety-relevant fault class, the braking system shall define and implement a safe state or degraded state that minimizes unreasonable risk.[web:16][web:24][web:29]

#### **REQ-FS-006 – Freedom from Interference**

Safety-relevant braking software and hardware elements shall be protected from interference by lower-integrity functions sharing computational or communication resources.[web:17][web:22]

#### **REQ-FS-007 – Monitoring and Fault Reaction Time**

Safety mechanisms shall detect and react to faults within time intervals consistent with the fault-tolerant time interval derived during safety analysis.[web:16][web:22]

## 12. Performance Requirements

### REQ-PF-001 – Stopping Performance

The braking system shall meet vehicle-category-specific stopping distance and deceleration performance requirements under prescribed test conditions.[web:6][web:7][web:11]

### REQ-PF-002 – Mean Fully Developed Deceleration

Where required by applicable standards, the braking system shall achieve minimum mean fully developed deceleration values during performance tests.[web:11][web:6]

### REQ-PF-003 – Straight-Line Stability

The braking system shall maintain straight-line stability during high deceleration events on uniform and split-friction surfaces within defined controllability limits.[web:6][web:30]

### REQ-PF-004 – Cornering Stability Support

When integrated stability functions are enabled, braking interventions shall support controllable vehicle behavior during braking in cornering scenarios.[web:30][web:21]

### REQ-PF-005 – Fade and Recovery

The braking system shall retain required minimum performance during fade testing and shall recover braking effectiveness after cooling within specified limits.[web:6][web:9]

### REQ-PF-006 – Power Supply Robustness for BBW

Brake-by-wire functions shall maintain required operating capability across specified supply voltage ranges and defined transient disturbances.[web:16][web:22]

## 13. Interface Requirements

### REQ-IF-001 – Vehicle Network Interface

Electronically controlled braking functions shall exchange status, fault, and torque or deceleration coordination signals with other vehicle controllers over defined communication interfaces.[file:32][web:30]

### REQ-IF-002 – ADAS Interface

The braking system shall support authenticated and validated brake requests from ADAS functions such as AEB when those features are part of the vehicle item definition.[web:12][web:29]

### REQ-IF-003 – Diagnostic Interface

The braking controller shall support service diagnostic access to fault memory, status variables, and calibration-relevant information subject to vehicle cybersecurity and service constraints.[file:32][web:16]

## 14. Requirement Relationships

The following requirement relationships are part of the specification and shall be maintained as explicit engineering traceability within the requirements set.[file:32][file:33]

- REQ-SYS-001 is refined by REQ-LG-001, REQ-BBW-003, REQ-PF-001, and REQ-PF-002.
- REQ-SYS-002 is refined by REQ-LG-022, REQ-EPB-001, and REQ-PF-001.

- REQ-SYS-003 is refined by REQ-ABS-002, REQ-ESC-001, and REQ-BBW-004.
- REQ-SYS-004 is refined by REQ-LG-002, REQ-LG-006, REQ-BBW-006, and REQ-BBW-008.
- REQ-SYS-005 is refined by REQ-FS-001 through REQ-FS-007.
- REQ-ABS-003 supports REQ-DIAG-001, REQ-DIAG-002, and REQ-FS-005.
- REQ-ABS-004 depends on REQ-ABS-003 and supports REQ-SYS-004.
- REQ-ESC-003 depends on REQ-DIAG-006 and supports REQ-SYS-003.
- REQ-BBW-001 is refined by REQ-BBW-020 and supports REQ-BBW-022.
- REQ-BBW-003 is refined by REQ-BBW-021 and supports REQ-PF-001.
- REQ-BBW-006 supports REQ-FS-004, REQ-FS-005, and REQ-SYS-004.
- REQ-BBW-023 is complemented by REQ-BBW-024 and supports REQ-PF-001 and REQ-PF-004.
- REQ-DIAG-003 and REQ-DIAG-004 support REQ-FS-007.
- REQ-FS-005 constrains REQ-ABS-004, REQ-ESC-003, and REQ-BBW-008.
- REQ-PF-006 constrains all brake-by-wire requirements from REQ-BBW-001 through REQ-BBW-024.

## 15. Comparative Summary

| Topic                     | Legacy Hydraulic                                                                                           | Brake-by-Wire One-Box                                                                                                                    |
|---------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Brake demand path         | Mechanical pedal plus hydraulic pressure generation.[web:13][web:14]                                       | Electrical demand sensing plus electro-hydraulic actuation.[web:18][web:23][web:28]                                                      |
| Assistance method         | Vacuum or hydraulic booster.[web:14][web:15]                                                               | Integrated electro-hydraulic pressure generation.[web:18][web:21]                                                                        |
| Base failure focus        | Hydraulic leakage, booster loss, mechanical retention, sensor faults in adjunct functions.[web:25][web:15] | Sensor plausibility, ECU faults, power supply robustness, pressure generation faults, fail-operational behavior.[web:16][web:23][web:25] |
| Regenerative coordination | Not typically applicable in conventional architecture.[web:6]                                              | Explicit friction/regen blending required in electrified platforms.[web:18][web:28]                                                      |
| Safety complexity         | Lower E/E coupling in base service brake path.[web:25]                                                     | High E/E dependency requiring stronger functional safety architecture.[web:16][web:22]                                                   |

## 16. Document Constraints

All requirement IDs in this specification are synthetic and intended for engineering, validation, and information-retrieval experimentation. They are realistic in style but are not lifted from any OEM proprietary specification.[file:32][file:33]